

KYLE MALONEY

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EDUCATION

Iowa State University — Ames, IA

May 2026

Bachelor of Science in Software Engineering, Minor in Cybersecurity

Relevant Coursework: Cloud Computing, Network Security, Cryptography, Operating Systems, Data Structures & Algorithms, Software Design

TECHNICAL SKILLS

Cloud & Infrastructure: AWS, Azure, Google Cloud (GCP), Terraform (IaC), Docker, Kubernetes (MicroK8s), container orchestration, Linux server administration

CI/CD & Monitoring: GitHub Actions, GitHub Container Registry (GHCR), Keel, Prometheus, Grafana, CloudWatch, Git, Bash scripting

Networking & Security: Cloudflare Tunnel (DNS, proxying, TLS), AdGuard Home, network configuration, Wireshark, Nmap, applied cryptography, Terrascan

Languages & Databases: Python, Bash, SQL, Java, C, C++, JavaScript/TypeScript, MySQL, SQLite, PostgreSQL, NoSQL

INFRASTRUCTURE EXPERIENCE

Self-Hosted Homelab — Kubernetes Application & Service Platform

Dec 2023 – Present

- Design, deploy, and maintain a self-managed MicroK8s (Kubernetes) cluster on Linux hosting production-style services: a media server, network-attached storage (NAS), photo management, a file manager, and public websites including my portfolio (thekyle.net) and blog.
- Built CI/CD pipelines with GitHub Actions and Keel that build the portfolio site and blog on every push and automatically roll updated pods out to the cluster.
- Expose internet-facing services through Cloudflare Tunnel, handling public DNS, proxying, and TLS with zero open inbound ports, and run internal DNS with AdGuard Home for network-wide ad and threat filtering.
- Monitor infrastructure health with Prometheus metrics and Grafana dashboards, and handle all Linux administration, networking, troubleshooting, and recovery from hardware and service failures.

CLOUD & DEVOPS PROJECTS

Terraform-Provisioned AWS Deployment ([Wiki-Guesser](#))

Jul 2026 – Present

- **Authored a full Terraform IaC stack** (VPC, EC2, IAM/OIDC, SSM, Route 53) for a right-sized, single-instance AWS deployment. Skipped the usual ALB, autoscaling, and RDS defaults since the app's architecture is stateful, which cut projected hosting cost from ~\$60/mo to ~\$20/mo.
- **Built a keyless CI/CD pipeline:** GitHub Actions assumes an AWS role via OIDC (no long-lived credentials, no SSH exposure) scoped to trigger a single SSM deploy document, with a separate PR-gated CI suite so main stays releasable at all times.
- **Set up monitoring:** CloudWatch alarms, SNS paging, and a Route 53 health check catch both infrastructure and application-level failures, not just server uptime.

Proxmox-Hosted Network Simulation & Model Deployment (GridSAFE)

Aug 2025 – May 2026

- **Stood up a simulated power-grid network on a self-hosted Proxmox cluster** to generate realistic traffic and injected-attack data for training and evaluating a detection model.
- **Trained an XGBoost anomaly-detection model that caught 80% of injected attacks in testing, then deployed it behind a REST API** so the monitoring dashboard could query live detection results.
- **Built a dashboard to launch simulated attacks and visualize detection results in real time**, collaborating with a 5-person multidisciplinary team across the full project lifecycle: network design, data generation, model training, and validation.

Secure Vault — Zero-Knowledge Encrypted Cloud Storage on AWS

Feb 2026 – May 2026

- Designed the cloud architecture for a zero-knowledge storage app: a Flask API backed by AWS S3 (boto3) that stores only client-side AES-256-GCM ciphertext, so it stays unreadable even if the provider is breached.
- Implemented envelope encryption with per-file wrapped data keys, enabling master-password rotation in a single atomic transaction without touching a byte of ciphertext in the cloud.

EXPERIENCE

Dining Hall Supervisor — Iowa State University Dining, Ames, IA

Oct 2022 – Feb 2026

- Supervised a team of student staff during high-volume dining service, coordinating daily operations and resolving service issues in real time while upholding health, safety, and quality standards.
- Created individualized weekly plans for students across multiple workstations to accomplish tasks efficiently and effectively.